

Post-Quantum Roadmap RFC for Terra Classic

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1 Scope Note (Formal)

This RFC bundle defines direction, decision logic, and validation paths for the Terra Classic PQ roadmap. It is intentionally not an implementation plan, delivery schedule, or resource commitment.

RFC stands for “Request for Comments.” In this context, the document is published to proactively gather feedback from all relevant Terra Classic stakeholders in order to improve scope clarity, content quality, readability, and practical feasibility before final freeze decisions are made.

2 Executive Summary (Informative)

2.1 Context and Purpose

Terra Classic currently uses cryptographic signatures in multiple security-critical areas. These signatures ensure that only authorized actors can execute valid actions and that the chain maintains a shared, verifiable state. Post-quantum cryptography (PQ, meaning cryptographic methods designed with future quantum-computing risks in mind) becomes relevant because sufficiently capable quantum computers may eventually attack and break classical signature algorithms that are widely used today.

This RFC provides a structured orientation framework for that transition. It defines how migration paths are identified, evaluated, and prioritized, and it describes the framework for audit gates, governance, and feedback up to RFC freeze. At the same time, this version is intentionally not a detailed technical specification and not a release, resource, or timeline commitment.

2.2 Goal in Plain Language

The strategic goal is to quantum-harden Terra Classic in a staged and auditable way, so the chain can continue secure operation even as long-term cryptographic risks evolve.

In simple terms: Terra Classic wants to prepare before cryptographic break risk becomes urgent, instead of reacting after ecosystem damage. The roadmap therefore focuses first on the highest-risk path (consensus), then on transaction signing for users/integrators, and finally on a durable Terra-Classic-native client ecosystem.

2.3 Document Structure

In the Terra Classic stack, signatures are used across several areas. In this document, these areas are called paths. The RFC first classifies these paths at a high level, then systematically defines their boundaries and criticality in the following chapters. Based on that, prioritization is derived and justified transparently.

After the executive summary, the document covers the current state and key terminology. This is followed by target states and options, then a phased roadmap. Afterward, audit gates, live-chain migration paths, and governance/feedback process are described. The document concludes with a decision log, where closed and open points are tracked transparently.

2.4 Risks, Impacts, and Benefits

2.4.1 If the roadmap is implemented

- Benefits: reduced long-term cryptographic exposure, earlier security option for advanced actors, stronger strategic independence from external client release cycles.
- Risks: migration complexity in consensus-critical paths, temporary operational overhead, and IBC continuity risk due to external counterparty dependency.
- Operational impact: larger signatures and additional verification logic may change resource and performance profiles; impacts must be measured and audited before production transitions.

2.4.2 If the roadmap is not implemented

- Risk concentration remains on classical cryptography over longer horizon.
- Terra Classic becomes more dependent on external timing for future emergency migration pressure.
- Delayed preparation can force compressed, higher-risk transitions later under less favorable conditions.

2.4.3 Positioning versus other chains

- Advantage potential: earlier preparedness and governance-visible audit discipline can improve trust for security-sensitive operators.
- Constraint: advantage depends on execution quality and ecosystem coordination, especially where interchain dependencies exist.

2.5 How Development and Rollout Work (Layman Terms)

- The roadmap is not “one giant overnight switch.”
- Work is split into packages and phases, each with explicit audit checks before the next risk step.
- New components are first built and tested in controlled test environments.
- Only after successful testing and gate approval can production migration decisions be made.
- For IBC-related changes, counterparties must also be coordinated because some required upgrades happen outside Terra Classic.
- If external readiness is partial, predefined fallback operating modes are used instead of forced unsafe cutover.

2.6 Reading Logic and Boundaries

The RFC is split into two chapter categories: Informative and Formal. Informative chapters explain motivation, structure, and decision logic in accessible language. Formal chapters deepen path definitions, risks, validation logic, and decision options for technical and governance decisions.

This formal/informative split is meant as orientation, not as exclusion. Non-technical readers are explicitly encouraged to read formal chapters as well; these chapters are intended to remain readable even when they are technically precise.

This executive summary does not pre-empt technical detail decisions and does not define a fine-grained implementation sequence. It serves as reading guidance. Detailed argumentation, assessments, and decision proposals follow in the subsequent chapters.

3 Cryptographic Signatures Explained Simply (Informative)

3.1 What Signatures Are About

A cryptographic signature is digital proof that a message actually comes from the person or system named as sender, and that the content was not modified unnoticed during transit. In practical terms, anyone verifying a signed message can check whether sender and content match.

In a blockchain system, this proof is central. Validators, wallets, and applications continuously make decisions based on signed data. If this proof is no longer reliable, technical uncertainty quickly becomes an operational and trust problem.

Without signatures, any party could claim that a message came from someone else. With signatures, it can be verified whether a message is authorized. This protects not only individual transactions but also the shared reliability and trustworthiness of the network.

In practice, complex cryptographic methods are used for this. These methods are much more sophisticated than simple teaching examples and are specifically designed to make forgery practically infeasible.

3.2 Asymmetric Signatures

In blockchain contexts, signatures are typically based on asymmetric cryptography. This means one party has two keys: a private key and a public key. The private key remains secret and is never shared. The public key is derived from the private key and can be distributed openly in the network.

The key point is that this derivation is designed not to be practically reversible. The private key should not be derivable from the public key. At the same time, the mathematical relation between both keys remains strong enough such that it allows signature verification to be computationally feasible.

3.3 Alice/Bob Positive Case

Alice sends a message to Bob and signs it with her private key. The signature is transmitted together with the message. Bob can then verify whether message and signature match and whether the message corresponds to Alice's public key.

Intuitively, the signature can be viewed as a special check value derived from two parts: the message content and Alice's private key. This links identity and integrity. Identity is expressed by key binding, integrity by message binding. If either part changes, verification fails.

Bob verifies signature, message and Alice's public key together. If verification succeeds, Bob can assume the message came from Alice and was not altered since it was signed. This is the normal case: signature verification creates binding between sender, content, and recipient.

In the negative case, an attacker tries to position themselves between Alice and Bob and impersonate Alice. If signatures were no longer secure or could be forged, Bob might accept a manipulated message as authentic. The result would not just be an error in one message, but a fundamental loss of trust in network communication.

3.4 Bridge to the PQ Topic

The core point of this RFC is not to explain specific algorithms in full detail, but to make the security assumptions behind real signature schemes explicit. Classical schemes are currently considered practically secure with typical parameters. With sufficiently capable quantum computers, this assumption may change for certain schemes.

This is exactly why Terra Classic components that rely on signature verification need a structured migration to new schemes: post-quantum schemes (PQ).

4 Current State and Terminology (Informative)

4.1 Which Signature Paths This RFC Covers

Before describing the status quo, this chapter defines the signature paths covered by the rest of the RFC. In this context, a path is not a single code section but a connected technical and operational domain in which signatures are created, transmitted, verified, and used as a basis for decisions.

For Terra Classic, three perspectives are most relevant. First, the consensus path, where validator signatures secure shared chain truth directly. Second, the wallet/tx path, where user transactions are signed and validated in backend flows. Third, the user-facing client path, where signing workflows are visible and operable for end users. In this chapter, the status-quo focus is on the two technically critical core paths, consensus and wallet/tx; the client path is deepened later in target states and roadmap phases.

This boundary definition is important because these paths have different risk profiles and migration effort. A failure in the consensus path has systemic impact on chain stability, while failures in the wallet/tx path more often appear as integration, compatibility, or usability problems. The RFC therefore does not discuss “signatures” in the abstract, but signatures in their specific operating context.

4.2 Status Quo of the Consensus Path

In the consensus path, signatures secure block production and finality between validators. Signature verification in vote-, proposal-, and commit-related flows is therefore part of the network’s security-critical core function. If verification in this area is no longer consistent or deterministic, this affects not only individual participants but potentially the stability of the whole system.

In the current Terra Classic context, **Ed25519** is the relevant scheme in the consensus path. The operational scope includes not just verification code, but also connected components for validator operations, key custody, and signature provision. This yields a clear baseline for the RFC: consensus-side signature changes must always be evaluated through safety, liveness, upgrade determinism, and operational stability. For this reason, the consensus path is treated as a high-criticality path in the roadmap.

4.3 Status Quo of the Wallet/Tx Path

In the wallet/tx path, signatures are created and validated for user transactions. This path is strongly ecosystem-shaped: wallets, custody environments, exchanges, APIs, SDKs, and integrations must work consistently in practice. Signature verification is security-relevant here as well, but impacts often first appear in compatibility, user flow, and support overhead.

In the current Terra Classic context, **secp256k1** is the relevant scheme in the wallet/tx path. The baseline is therefore different from the consensus path. In the wallet/tx domain, cryptographic correctness alone is not sufficient; it also matters how broadly new signing behavior can be integrated into existing toolchains and interfaces. For the RFC, this means this path always requires integration and operations argumentation alongside security argumentation.

4.4 Why This Baseline Is Decisive for the Roadmap

The roadmap is built on this status-quo analysis. Only clear path separation allows priority, risk, and migration logic to be justified transparently. Without this separation, technical decisions would easily become over-generalized and harder to compare.

This chapter therefore provides the shared foundation for following chapters: target states and options build on the same path boundaries, audit gates follow the same risk profiles, and governance decisions are evaluated along the same structure.

4.5 Terms for Following Chapters

- **Path** - In this RFC, a path is a connected technical and operational domain where signatures are created, transmitted, and verified.
- **Consensus path** - The consensus path includes signing and verification flows that secure shared chain truth directly, especially coordinated validation and production of blocks.
- **Wallet/tx path** - The wallet/tx path includes user-side transaction signatures and their validation in backend and protocol flows. These signatures ensure individual transactions are authorized by individual users.
- **Client path** - The client path is the user-facing layer where signature processes are visible and operable in wallets and interfaces. On the client side, signature generation is more central than verification.
- **Cryptographic signature** - A cryptographic signature is digital proof binding sender and content.
- **Signature verification** - Signature verification is the check that message, signature, and public key are logically consistent.
- **Private key** - The private key is a party's secret key and is used to produce signatures.
- **Public key** - The public key is the distributable counterpart to the private key and is used for verification.
- **Authenticity** - Authenticity means a message truly comes from the claimed sender.
- **Integrity** - Integrity means a message was not altered unnoticed after signing.
- **Validator** - A validator is a network participant that creates and verifies signatures in the consensus process.
- **Voting Power (VP)** - Voting power is a validator's consensus weight.
- **Safety** - Safety in consensus means the network does not accept contradictory valid states.
- **Liveness** - Liveness means the network continues producing blocks under valid operating assumptions and does not halt permanently.
- **Upgrade determinism** - Upgrade determinism means a protocol or software switch yields identical results for all participants under identical conditions.
- **Cutover** - A cutover is a planned switch point between two operating modes.
- **Audit gate** - An audit gate is a mandatory validation and release checkpoint before moving to the next phase.
- **In-place upgrade** - An in-place upgrade is a migration path where the existing live chain is continued in a controlled way while being technically switched over.
- **Re-genesis** - Re-genesis is the fallback path if in-place upgrade is not viable.
- **Key binding** - Key binding is the traceable assignment of a new key to an existing validator identity.
- **CometBFT** - CometBFT is a Terra Classic software component where core consensus and validator signing flows are anchored.
- **Cosmos SDK** - Cosmos SDK is the modular application framework Terra Classic uses for core chain logic, transaction handling, and module composition.
- **wasmd** - wasmd is the CosmWasm integration layer used by Terra Classic to run and manage smart contracts in the chain runtime.
- **IBC** - IBC (Inter-Blockchain Communication) is the protocol stack used to exchange packets and verify state proofs across connected chains.
- **ibc-go** - ibc-go is the Go implementation of IBC modules and middleware used by SDK-based chains, including Terra Classic.
- **Ante** - In SDK context, Ante is the transaction pre-check layer before execution.
- **RFC** - RFC stands for "Request for Comments" and in this context means a structured discussion and decision document that makes direction, options, risks, and open points transparent before binding delivery decisions are made.

- **Binding** - Binding means formally mandatory within this RFC/governance context: once marked binding, it is the required baseline until changed through the defined revision/change-control process.
- **Delivery** - Delivery means practical implementation and operational rollout work (code, testing, migration preparation, operations execution), as distinct from RFC discussion and direction setting.
- **Decision** - A decision is a formally documented choice between options, including rationale and status (open, closed, deferred) in the Decision Log.
- **Ed25519** - Ed25519 is the relevant signature scheme in the current Terra Classic consensus path.
- **secp256k1** - secp256k1 is the relevant signature scheme in the current Terra Classic wallet/tx path.
- **PQ (Post-Quantum)** - PQ denotes cryptographic methods intended to be more robust against threat models with capable quantum computers than many widely used classical methods.
- **ML-DSA** - ML-DSA is defined in this plan as the target profile for the consensus path.

5 Target States and Options (Formal)

5.1 Consensus Target State

For the consensus path, the target state is a clear PQ-only operation using ML-DSA from a defined switch day (Day X). This means that after cutover, the consensus path should not run with two signature regimes in parallel as a permanent mode, but should have one unambiguous end state.

To make the target state traceable, the RFC considers two options. Option A is a direct cutover into PQ-only consensus mode from Day X. Option B is a hybrid transition mode in which classical and new signature methods are accepted in parallel for a limited time in the consensus path.

The hybrid model is not adopted as the consensus target state because it expands rule complexity in the most security-critical path and therefore increases complexity, attack surface, and coordination risk. In addition, hybrid operation creates extra testing, audit, and operational overhead because multiple valid signature modes must be processed correctly and deterministically at the same time. The RFC therefore prioritizes a clear end state with unambiguous verification behavior.

5.2 Wallet/Tx Target State

In the wallet/tx path, the target state is a hybrid transition path rather than an early hard PQ-only cutover. Classical and PQ-resistant signatures are supported in parallel during a defined transition period, so adoption can be matched to security needs, technical maturity, and operational capability of each participant group.

The focus is therefore on orderly introduction, clear compatibility boundaries, and transparent migration logic for users and integrators. In the early phase, support for PQ-resistant methods in client and user-facing environments may still be patchy, while UX, tooling, and recovery processes mature only in later RFC phases.

This target state therefore prioritizes staged adoption: security-critical and technically advanced actors such as CEXs, custody providers, and infrastructure projects can use PQ signatures early in production and secure their systems earlier. Less technical retail participants can follow step by step as wallet ecosystem, standards, and usability mature.

5.3 PQ-Native Client Target State

The target state for PQ-native clients is the long-term buildout of a Terra-Classic-native wallet and user-facing client ecosystem with native PQ compatibility. Signature generation, signature verification, and key management should become available and practical for end users early, instead of waiting for external maturity cycles in the broader Cosmos or blockchain environment.

Buildout happens with active involvement of infrastructure actors already operating productively in the Terra Classic ecosystem and having proven operational maturity. This ensures that the rollout of PQ-compatible wallet systems does not depend on third parties, large external ecosystem actors, or externally set standards in a still dynamic PQ landscape.

The target state is that individual actors and retail participants within Terra Classic can participate in PQ-friendly signature methods as early as possible to protect their assets, without being blocked by potentially slow PQ adoption outside Terra Classic.

5.4 Intentionally Open Points

Despite clear core direction decisions, this chapter intentionally keeps some points open that should only be finalized through further technical elaboration and stakeholder alignment. This includes concrete integration details per ecosystem component, exact transition conditions for individual sub-paths, and the granularity of rollout and communication logic.

All open points are tracked explicitly in the Decision Log and marked as open, closed, or deferred. This keeps transparent which target states are already binding and where decision work remains.

6 Roadmap Phases (Formal)

6.1 Phase Omega: RFC Freeze and Start Authorization (Preceding)

Phase Omega is the upstream completion of RFC drafting before any technical implementation starts. It is not a delivery plan and not an implementation phase, but the formal transition from “discussion and variants” to a “binding implementation baseline” (the frozen, governance-approved RFC reference that implementation work must follow unless formally revised). Without a completed Omega gate, implementation phases A through C do not begin.

The purpose of this phase is to establish a robust content baseline for all downstream technical decisions. Open conflicts are not left implicit; they are explicitly decided, deferred, or rejected. At the same time, this phase defines which parts of the RFC are frozen and under which conditions post-freeze changes are allowed.

6.1.1 Work Package O0: RFC Launch and Formal Comment Solicitation

O0 is the formal start package for this RFC process. The RFC was initially drafted by the author in single-author mode and is then presented and published through a Terra Classic governance proposal as the formal kickoff for stakeholder review.

The proposal explicitly asks whether Terra Classic stakeholders should formally enter the RFC comment process and requests relevant stakeholder groups (see Chapter 9, Stakeholder Groups) to submit comments through the defined channels (see Chapter 11, Submission Channel).

The mandatory result of O0 is a published kickoff baseline with explicit governance-level invitation to comment, clear stakeholder addressing, and active comment intake.

6.1.2 Work Package O1: Consolidation and Decision Closure

O1 formally closes the RFC feedback cycles. Comments and opposing positions from stakeholder rounds are classified and transferred into document state with reasoned decisions. The Decision Log is maintained as a binding reference, including status per point (closed, deferred, rejected) and traceable rationale.

The mandatory result of O1 is a content-consistent RFC state without hidden conflicts in core questions. What remains open is explicitly marked as deferred and assigned to a clear follow-up process. O1 is formally closed through a dedicated governance proposal that confirms decision closure and the consolidated RFC baseline.

6.1.3 Work Package O2: Phase and Gate Consistency Review

O2 ensures consistency of implementation logic across chapter boundaries. It reviews target states, scope boundaries, audit gates, and exit criteria in phases A through C, as well as compatibility with governance and migration chapters.

The mandatory result of O2 is a documented consistency review showing clearly that there are no blocking conflicts between Informative and Formal chapters, and no silent scope shifts being carried into implementation.

6.1.4 Work Package O3: Governance Freeze and Reference Publication

O3 executes the formal governance event for freeze (for example proposal and vote) and converts the consolidated state into a frozen reference version. This version is the binding basis for implementation in phases A through C.

Part of O3 is defining post-freeze change rules: which changes count as editorial only, and which require a new RFC revision and renewed governance approval.

6.1.5 Dependencies and Order

Omega packages run in clear sequence. O1 depends on a completed kickoff and formal comment solicitation from O0. O2 depends on formally approved O1 closure and the resulting consolidated baseline. O3 depends on a passed consistency review from O2. This sequence ensures unresolved conflicts do not enter a formal freeze decision.

6.1.6 Preliminary Exit Criteria (Omega Go/No-Go)

- O0 publishes the RFC kickoff through governance and formally opens stakeholder comment intake through defined channels.
- O1 is complete, the Decision Log is finally consolidated including explicitly marked deferrals, and formal closure is approved through governance proposal.
- O2 documents and confirms consistency between Informative and Formal chapters and between phase and gate logic.
- O3 is formally approved and published as frozen reference version.
- Post-freeze change rules are bindingly documented.

6.1.7 Out of Scope in Phase Omega

- Implementation work in code.
- Commitments on timeline, budget, or resources.
- Premature detailed technical design decisions that belong to implementation phases A through C.

6.2 Phase A: Consensus Path PQ-Enabled

Phase A establishes the technical decision and implementation foundation for a PQ-enabled consensus path. The RFC intentionally works in downstream chapters with the premise that a PQ-resistant CometBFT fork is fundamentally feasible. At the same time, this assumption is not taken for granted in Phase A; it must be explicitly validated. The feasibility study acts as an early invalidation mechanism: if it is negative, the RFC in its current form is stopped and moved into an alternative roadmap.

Implementation in Phase A is split into clearly bounded work packages that can be delivered by internal and/or external software development providers. The focus is secure transition of the consensus path. End-user wallet/tx rollout and broad user-client ecosystem expansion are not part of this phase.

6.2.1 Work Package A1: CometBFT Fork Feasibility Study (Gate 0)

A1 robustly tests whether a PQ-resistant consensus path based on a CometBFT fork is technically, security-wise, and operationally viable for Terra Classic. The study covers not only the fork core but the full dependency chain of the consensus stack: primarily CometBFT, additional interfaces extending into Cosmos SDK, and potential effects on smart-contract layer integrations including wasmd-near interfaces.

The mandatory result of A1 is a documented Go/No-Go evaluation with clear no-go criteria, risk classes, and invalidation signals. A1 also includes structured decomposition of follow-up work into consistent packages for prototype, audit, and migration.

6.2.2 Work Package A2: PQ Prototype on Independent Testnet

A2 delivers a working prototype of a PQ-resistant CometBFT fork using ML-DSA. The prototype must prove technical integration in a realistic stack and therefore includes required dependencies extending into Cosmos SDK and wasmd.

The prototype runs on a genesis testnet independent of `rebel-2`. Migration of an existing live chain is not the focus of this package. The focus is functional behavior, integration consistency, reproducible operation, and robust measurement data for next decisions.

6.2.3 Work Package A3: Audit Gate for Prototype (Gate 1)

A3 is the formal security and quality gate between prototype and migration development. The prototype is independently reviewed before migration components for a production path are built.

Audit focus includes safety/liveness impact, cryptographic correctness of ML-DSA integration, deterministic behavior in critical consensus paths, and failure modes under load and malformed inputs. The output is a documented Go/No-Go decision for transition to A4.

6.2.4 Work Package A4: Migration Components for Consensus Path

A4 develops all technical components and operational strategies required for orderly transition to the consensus target state. This includes upgrade and migration handlers, mechanisms to register PQ signatures or PQ keys for existing validators, and formal rule sets for cutover, activation, and safe post-switch operation.

This package ensures the transition is not only technically possible but operable within production-grade governance and operations logic. The focus is deterministic behavior in the upgrade window and unambiguous rules for valid signatures before, during, and after cutover.

6.2.5 Work Package A5: IBC Compatibility and Counterparty Migration Preparation

A5 specifies and implements all IBC-relevant migration components required to keep interchain operation viable across consensus cutover. This includes explicit compatibility strategy for existing `07-tendermint` clients, introduction path for a successor client type where required, relayer runbooks, and chain-by-chain counterparty upgrade coordination requirements.

The package must produce a deterministic operational plan for the cutover window: how client updates are sequenced, how channel disruption risk is limited, what fallback and recovery procedures apply, and which readiness evidence must be reached before productive switch.

Critical risk premise: the decisive client upgrades are executed on counterparty chains, not on Terra Classic. Counterparty governance decisions, release cycles, and operational priorities are external dependencies and cannot be enforced by Terra Classic. Therefore participation and timeline alignment are explicitly non-assured and must be treated as major Go/No-Go risk in migration governance.

After the technical implementation profile for the PQ signing client path is sufficiently specified, A5 includes a mandatory active engagement step by roadmap executors: direct talks with relevant actors on each counterparty chain (governance participants, core maintainers, relayer operators, and major ecosystem operators) to negotiate upgrade feasibility, sequencing, and commitment level.

Operational concentration risk is explicitly in scope: at the time of this RFC revision, `LuncGoblins` is the currently known paid IBC relayer operator for Terra Classic. A5 must therefore include a relayer-readiness and continuity plan that addresses single-operator dependency risk during migration.

6.2.6 Work Package A6: Audit Gate for Migration Components (Gate 2)

A6 is the final audit gate before productive consensus-path transition. It reviews correctness and tamper resistance of registration and migration mechanics, consistency of cutover rules, and secure deterministic post-switch startup.

The result is a robust Go/No-Go recommendation for the production migration path. Critical findings block transition to downstream rollout and governance steps until fixed and successfully re-audited.

6.2.7 Dependencies and Order

Packages are intentionally sequential. A2 requires a positive A1 result. A4 requires a passed audit gate from A3. A5 depends on migration logic from A4 and must complete before final gate review. A5 additionally depends on external counterparty readiness signals that Terra Classic cannot unilaterally control. A6 is mandatory before any production migration decision. This sequence prevents unverified assumptions from early development from leaking into live migration.

6.2.8 Preliminary Exit Criteria (Phase A Go/No-Go)

- A1 is complete and includes an explicit continuation decision based on traceable no-go criteria.
- A2 delivers a reproducibly running prototype on an independent genesis testnet.
- A3 passes without critical blockers for entering migration development.
- A4 fully specifies and implements migration components and cutover logic.
- A5 delivers a validated IBC compatibility and counterparty migration plan with explicit cutover readiness signals and documented counterparty participation status.
- A5 explicitly documents residual external dependency risk where counterparty participation is missing, uncertain, or time-unbounded.
- A5 documents outreach execution and outcomes from direct counterparty-chain talks after technical client-path clarification, including chain-specific blockers and commitment signals.
- A5 defines fallback operating modes for partial or delayed counterparty adoption, including routing-impact communication and recovery sequencing.
- A5 includes a relay-readiness and continuity plan that explicitly addresses the current LuncGoblins single-operator dependency.
- A5 includes a counterparty test campaign with rehearsal scenarios and documented pass/fail evidence.
- A5 produces governance package templates for counterparty-side proposal and operator communication flows.
- A6 passes without critical blockers for production migration path.

6.2.9 Out of Scope in Phase A

- Wallet/tx rollout for end users.
- Buildout of broad PQ-native user-client ecosystem.
- Delivery details such as owners, dates, or budgets.

6.3 Phase B: Wallet/Tx Path PQ-Enabled

Phase B creates the technical implementation foundation for the hybrid signature path in wallet/tx. The goal is not an early hard PQ-only cutover, but an orderly parallel capability of classical signature verification and ML-DSA in the production-relevant transaction path.

The baseline differs from Phase A: Cosmos SDK already provides a fundamentally suitable framework for alternative signature and verification mechanisms via abstract account/key types, flexible data structures, customizable tx types and extensions, and a programmable ante handler. Even though secp256k1 is predominant in current transaction signatures, Phase B therefore does not need a separate upstream feasibility gate. It starts with coordinated requirements and architecture planning.

6.3.1 Work Package B1: Coordinated Planning Phase (Requirements)

B1 consolidates functional and technical requirements needed to reach the hybrid signature target state. Core focus is target architecture for ante handler and signature verification, plus required extensions in key/account types, tx extensions, and adjacent interfaces.

The mandatory result of B1 is an aligned implementation and migration work backlog. This includes clear compatibility boundaries, error and fallback behavior, and a robust test approach for subsequent prototype operation.

6.3.2 Work Package B2: Development, Component Transition, and Prototype Test Run

B2 implements hybrid validation including ML-DSA support across all affected wallet/tx components along the real transaction flow. The goal is not isolated integration, but consistent transition of dependent components in combined operation.

As technical proof, an end-to-end prototype is operated on a dedicated testnet. This package is successful when hybrid signature validation runs stably and reproducibly with traceable operational behavior.

6.3.3 Work Package B3: Audit Gate for Hybrid Wallet/Tx Path

B3 is the mandatory security and quality gate after implementation and prototype operation. It reviews correctness of signature validation, security properties in ante and verification paths, and robustness under load and failure scenarios.

The result is a documented Go/No-Go decision for transition to migration elaboration. Critical findings block progression until remediation and renewed review.

6.3.4 Work Package B4: Migration Components for ML-DSA Introduction

B4 specifies and implements migration components for orderly ML-DSA introduction in wallet/tx. This includes introduction strategy, activation logic, compatibility windows, and operational transition for hybrid mode.

B4 intentionally has no separate additional mandatory audit gate as an independent milestone. Depending on observed complexity, B3 and B4 may be merged or tightly coupled, provided shared Go/No-Go logic remains documented.

6.3.5 Dependencies and Order

Packages are sequential. B2 requires a completed B1 requirements package. B4 requires a passed audit gate from B3 unless B3 and B4 are intentionally run as one combined block. This order ensures migration elaboration is based on validated, not merely implemented, mechanisms.

6.3.6 Preliminary Exit Criteria (Phase B Go/No-Go)

- B1 is complete and provides aligned requirements including architecture and test baseline.
- B2 delivers reproducible end-to-end prototype proof on a dedicated testnet.
- B3 passes without critical blockers and allows transition into migration elaboration.
- B4 is implementation-ready; if B3/B4 are merged, a shared documented Go/No-Go decision applies.

6.3.7 Out of Scope in Phase B

- Full rollout of a Terra-Classic-native user-client ecosystem.
- Delivery details such as owners, dates, or budgets.

6.4 Phase C: PQ-Native Clients

Phase C translates technical PQ capability from wallet/tx into a durable Terra-Classic-native user ecosystem. Unlike phases A and B, the structure here is intentionally not strictly sequential. Core work packages can start in parallel and be iteratively aligned.

The core objective is that Terra Classic does not remain dependent on third parties or external release cycles in the broader ecosystem for PQ-capable wallet, explorer, and user-facing infrastructure.

6.4.1 Work Package C1: Terra-Classic-Native Wallet/Explorer Stack

C1 includes requirement definition, procurement, and commissioning of an own wallet/explorer stack that fully reflects the hybrid and later expanded PQ-capable wallet/tx path.

Development may be performed by external providers, but is operated as Terra-Classic-owned infrastructure under a public/open license model. The goal is an independently operable, openly licensed, and traceably maintainable stack for retail and individual users.

6.4.2 Work Package C2: Public-Law Carrier Entity and Operational Ownership

C2 establishes a public-law entity that institutionally carries the stack. This entity defines requirements, commissions implementation, manages ownership, and is accountable for monitoring and hosting.

Scope includes not only web instances and backend nodes, but also ownership/control of critical digital assets such as DNS names, developer accounts, and publication channels (for example app stores, package registries, repositories). C2 also includes drafting a constitutional-style baseline relationship between this entity and Terra Classic governance module.

6.4.3 Supporting Work Package C3: External PQ Standards Monitoring and Alignment

C3 runs as a permanent parallel stream to C1 and C2. Its purpose is active contact and relationship management with relevant actors in Cosmos, Ethereum, and broader blockchain ecosystems to monitor PQ standardization early, identify divergence in time, and proactively align Terra Classic decisions.

This stream is security-strategically required: ML-DSA is used as current target algorithm, but without assuming mathematical finality against future cryptanalytic breakthroughs (quantum or classical). Terra Classic decisions must therefore remain revision-capable and responsive to external evidence growth.

6.4.4 Dependencies and Parallelization

C1 and C2 are fundamentally parallelizable and need not run as a rigid sequence. C3 accompanies both continuously. Where operational coupling emerges (for example release ownership, operational obligations, compliance constraints), synchronization points are documented as governance decisions rather than enforced as implicit ordering.

6.4.5 Preliminary Exit Criteria (Phase C Go/No-Go)

- C1: Requirements are approved, commissioning is completed, and a functioning publicly licensed wallet/explorer stack is demonstrable.
- C2: The carrier entity is formally established and demonstrably assumes ownership, operational responsibility, and asset control (DNS, developer accounts, distribution channels).
- C2: The baseline relationship between carrier entity and Terra Classic governance module is bindingly formulated and published.
- C3: A running external monitoring and alignment process is established, with documented contacts, review cycles, and feedback into technical decisions.

6.4.6 Out of Scope in Phase C

- Detailed planning of individual procurement/tender procedures at operational level.
- Full anticipation of all future PQ algorithm shifts; Phase C establishes institutional and technical adaptability for that.

7 Audit Gates as a Mandatory Component (Formal)

7.1 Purpose and Core Principle

Audit gates are the binding security and quality mechanism between roadmap phases. They prevent insufficiently validated assumptions from prototype or migration work from moving into live operation.

Core principle: without a passed gate, no recommendation is issued to proceed into the next risk-relevant step. Exceptions are only allowed as explicit governance risk acceptance and must be documented with rationale, residual risk, and remediation plan.

7.2 Audit Logic Across Phases

The RFC uses different gate types by phase: - Phase Omega: consistency and governance gate (content/process, no code security audit). - Phase A: two mandatory technical security gates (prototype and migration components) for consensus path. - Phase B: one mandatory technical security gate after prototype operation for hybrid wallet/tx path. - Phase C: no single rigid one-shot gate, but an ongoing assurance model for product, operations, and supply-chain risks.

This keeps gate strictness aligned with phase criticality: strictest in consensus, strict in wallet/tx, continuous operational assurance in client ecosystem.

7.3 Gate Matrix (Binding Minimum Logic)

1. Omega Gate: Phase Omega must be complete before A-C start.
2. A Gate 1 (after A2): audit of PQ consensus prototype as prerequisite for A4.
3. A Gate 2 (after A5): audit of consensus migration components as prerequisite for production migration decision.
4. B Gate (after B2): audit of hybrid wallet/tx path as prerequisite for B4.
5. B4 intentionally has no second separate mandatory gate; with high complexity, governance may require additional re-audit.
6. C Assurance: recurring review cycles instead of single release gate.

7.4 Mandatory Components of Every Audit Package

Each gate must include at least: - Audit objectives and explicit scope (including out-of-scope). - Audit type (internal, external, or combined) with role split. - Auditor input artifacts (code, architecture, test evidence, threat model, operations documentation). - Findings by criticality class. - Re-audit rules per criticality class. - Formal Go/No-Go recommendation with residual-risk assessment.

7.5 Finding Classes and Blocker Rules

Binding minimum classes: - Critical: direct blocker; no transition without fix and re-audit. - High: blocker for production-near or production steps; exception only via explicit governance risk acceptance. - Medium: no automatic blocker, but requires deadline and tracking. - Low/Info: documentation and backlog required, no gate blocker.

A gate counts as passed only when all Critical findings are closed and no uncontrolled High risk exposure remains.

7.6 Re-Audit and Change Rules

Re-audit is mandatory for: - Fixes for Critical or High findings. - Changes in cryptographic core paths (signing or verification logic). - Changes in cutover, migration, or key-binding mechanisms. - Changes affecting determinism, safety/liveness, or key management.

After a passed gate, relevant code and configuration changes must be documented against audit scope up to the next milestone, to prevent scope drift.

7.7 Mandatory Review Points in Consensus Path (Phase A)

- Safety/liveness properties of forked CometBFT under PQ integration.
- Correct verification in proposal, vote, and commit paths.
- Determinism across upgrade and cutover windows.
- DoS risks from larger signatures, changed verification costs, and load spikes.
- Hardening of PrivValidator/remote-signer interfaces.
- Correctness and tamper resistance of key binding (classical → PQ) and snapshot/activation logic.
- IBC continuity risks from consensus-signature transition, including dependency on counterparty client upgrades and non-participation scenarios.
- Relayer-readiness and continuity controls for migration, including concentration risk treatment for the currently known single paid Terra Classic relayer operator (LuncGoblins).
- Counterparty test-campaign evidence, including rehearsal outcomes for partial-adoption fallback modes.
- Negative tests (malformed input, boundary sizes, mixed-version scenarios).

7.8 Mandatory Review Points in Wallet/Tx Path (Phase B)

- Correct signature verification and ante-handler behavior in hybrid signature paths.
- Replay, malleability, and domain-separation checks.
- Gas/fee hardening and anti-spam/DoS behavior in hybrid mode.
- Correct codec, serialization, and relevant derivation rules.
- Key-management and recovery flows for users and integrators.
- Compatibility tests across introduction and migration windows.

7.9 Mandatory Review Points in Client/Operations Context (Phase C)

Phase C focuses long-term operation of a Terra-Classic-native wallet/explorer stack and the carrier structure. It requires continuous assurance on:

- Operations and hosting security (web, backend, node operations, monitoring).
- Supply-chain security and release integrity (build, distribution, signing, publisher accounts).
- Governance and ownership control of critical assets (DNS, repositories, package/app distribution).
- Incident, patch, and vulnerability response processes with traceable escalation paths.
- External PQ standards monitoring and technical decision feedback.

7.10 Documentation Duty per Gate

Each gate must produce an auditable decision dataset:

- Scope and version of reviewed artifacts.
- Findings list with status and ownership.
- Risk acceptances (if any) including deciding authority.
- Go/No-Go decision and approved next step.

This ensures the roadmap is not only implemented technically, but also governed reproducibly and traceably across its risk-critical transitions.

8 Live-Chain Migration Paths (Formal)

8.1 Positioning of This Chapter

This chapter is categorized as formal. It provides a more detailed view of the consensus cutover path and explains how emergency scenarios are evaluated if the primary migration path cannot be executed safely or in time.

It is not a delivery plan, but a technical orientation for transitioning from current consensus operation to PQ-based consensus operation.

8.2 Primary Path: In-Place Cutover

The primary migration path is a coordinated in-place upgrade with planned chain halt and subsequent restart on the forked, PQ-enabled consensus stack. This path prioritizes live-chain continuity and minimizes ecosystem disruption.

A prerequisite is that preparatory safety conditions from Phase A are satisfied, especially valid and robustly reviewed binding from existing validator identities to new PQ consensus keys and a documented IBC counterparty migration readiness assessment.

8.3 Core Element: Binding Existing Validators to New Consensus Keys

The security-critical core of cutover is unambiguous assignment of “existing validator -> new PQ consensus key.” Without this mapping, post-switch operation cannot verify which new key belongs to which previous validator identity.

Binding is handled as a prepared and traceable registration process: - Each existing validator registers its new PQ consensus key through the defined mechanism. - The process enforces uniqueness and prevents conflicting duplicate mappings. - A final deterministic snapshot of valid bindings is produced before cutover. - Cutover release requires sufficient registered voting power (per Phase A logic).

8.4 Consensus Cutover Sequence (Informative)

1. Activate binding window and run ongoing PQ consensus key registration.
2. Validate binding quality and reach required voting-power threshold.
3. Create and verify final binding snapshot.
4. Execute coordinated chain halt at defined switch point.
5. Deploy/start PQ-enabled consensus stack with approved snapshot.
6. Resume block production with PQ consensus keys.

8.5 Why No Direct Immediate Chain Halt Without Binding

An immediate halt and switch without prior binding is not a robust path because secure identity continuity for validators would be missing. This increases risk of misassignment, operational failure, and disputes during validator activation after restart.

8.6 Re-Genesis Assessment: Currently Not a Robust Fallback

For Terra Classic, re-genesis is currently not assessed as a practical fallback but as an operational dead end. For the existing chain state in the range of hundreds of gigabytes, there is still no robust proof that safe, reproducible genesis export plus stable import can be done in acceptable time.

Available experience reports indicate that even under extreme RAM and disk-I/O conditions, genesis import was aborted after long runtime (for example around 24 hours). Therefore there is currently no proof that re-genesis is a safe emergency strategy under realistic operating conditions.

Consequence for this RFC: migration must not depend on re-genesis as a reliable rescue path. Risk reduction must be achieved primarily inside in-place cutover itself (binding quality, gate discipline, deterministic snapshot, coordinated operational transition).

8.7 Decision Criteria and Emergency Principles

The decision on in-place cutover follows a strict Go/No-Go principle: - Safety before speed. - Determinism before ad-hoc workarounds. - Documented governance decision before operational shortcuts. - No implicit assumption of counterparty participation in required IBC client upgrades.

If a Go cannot be robustly justified, no unsafe cutover is forced. This includes cases where counterparty migration participation is missing, uncertain, or not aligned with cutover timing. Instead of switching to a currently unvalidated re-genesis path, hardening, re-audit, and renewed decision cycle are required.

8.8 Fallback Operating Modes for IBC

Because counterparty client upgrades are external dependencies, this RFC requires explicit fallback modes for partial adoption scenarios instead of binary assumptions.

Minimum fallback-mode coverage: - Mode 1 – Full continuity: counterparties required for target routing have completed client-side readiness; standard cutover sequence proceeds. - Mode 2 – Partial continuity: only a subset of counterparties is ready; operation proceeds with explicitly declared route limitations and prioritized relaying on viable paths. - Mode 3 – Protective restriction: readiness is insufficient for safe broad routing; IBC operation is temporarily restricted according to predefined safety policy while migration hardening and coordination continue.

For each mode, A5 artifacts must define operator actions, relayer posture, user-facing communication scope, and the condition for transitioning to the next mode.

9 Governance, Communication, and RFC Process (Formal)

9.1 Governance Decisions

This chapter defines how decisions across the RFC lifecycle are prepared, made, documented, and communicated. The goal is a traceable process combining technical quality, stakeholder participation, and formal decisionability.

In this RFC, governance does not mean operational delivery steering. It means content-level direction decisions, release logic, and transparent risk acceptance across phases Omega through C.

This chapter distinguishes two governance scopes to avoid ambiguity:

- **RFC Governance Authority** - the process-level authority for active RFC operations (editing flow, versioning flow, comment processing flow, and preparation of closure/freeze decisions).
- **Terra Classic Chain Governance** - the on-chain governance body that can observe the RFC process and decide formal proposals affecting RFC direction, closure, or freeze.

During the open RFC process, document editing and versioning are intentionally not operated as day-to-day chain-governance actions. This preserves operational agility while comments and decisions are still being incorporated. Chain governance remains fully entitled to question, challenge, or revise RFC decisions through formal proposals, but this is defined as an escalation path, not the default operating mode.

Binding principle: traceability is prioritized over speed, risk acceptance must always be explicit, no phase-critical continuation is allowed without satisfied gate logic, and post-freeze technical deviations are only permitted via formal change rules.

9.2 Stakeholder Groups

The following groups are central for feedback, impact, and practical feasibility:

- **Users:** end users whose asset protection, usability, and migration capability are central to wallet/client strategy.
- **Validators:** operators of consensus infrastructure, critical for key binding, cutover stability, and safe post-switch startup.
- **Wallet providers:** teams/products implementing signing, key management, and recovery flows for hybrid and later PQ-native operation.
- **Exchanges:** central trading/settlement actors with high security requirements and early integration relevance for PQ-capable signature paths.

- **Custody providers:** professional custodians with elevated responsibility for key control, operational security, and auditable processes.
- **Integrators:** infrastructure and product teams that must keep nodes, APIs, SDK-adjacent services, and operations processes compatible with new paths.
- **External infrastructure and standardization actors:** references from broader ecosystem (for example Cosmos/Ethereum-adjacent projects, security research, tooling providers) relevant for Phase C and ongoing PQ alignment.

9.3 Roles in the RFC Process

The process distinguishes functional roles:

- **Commenters** - submit RFC comments through the defined channels, reference affected chapters, and provide rationale for proposed changes or objections.
- **RFC Editors** - consolidate text contributions, maintain chapter consistency, and maintain versions and change logs.
- **Technical Maintainers** - assess technical coherence, risk assumptions, and gate readiness across chapters and phases.
- **RFC Governance Authority** - makes process-level release, closure, and freeze preparations and coordinates formal decision packages for governance steps.
- **Terra Classic Chain Governance** - retains final on-chain decision rights where formal proposals are required and can challenge or revise RFC direction when needed.
- **Auditors** - perform internal, external, or combined gate assessments as defined in Chapter 7.

Roles may overlap in persons, but documentation must still clearly separate who proposed, reviewed, and finally decided.

9.4 RFC Feedback Cycle

The RFC feedback process is continuous during the open RFC phase, not round-based. Incoming comments are reviewed on an ongoing basis, incorporated where accepted, and reflected in updated document versions and decision records.

In the default flow, each relevant comment can trigger a new document revision. Where appropriate, multiple comments may be consolidated into one combined revision, provided traceability per comment is preserved in the Comment Log and corresponding decision updates remain explicit in the Decision Log.

This model is intentionally lightweight and operationally flexible for a decentralized environment: no fixed feedback windows and no mandatory institutional review rounds are required as the default operating pattern.

9.5 Comment Submission and Review Process

Comments are submitted through an RFC mailing list (to-be-done, list details will be published before the next feedback round). The process is:

1. A commenter submits a comment to the mailing list with a clear subject, chapter reference, and rationale.
2. RFC editors review the comment for scope fit, clarity, and actionability.
3. Editors assign one of two outcomes: accepted or rejected.
4. Every reviewed comment is recorded in Chapter 11 (Comment Log) with status and short rationale.
5. Accepted comments usually result in a document change; the corresponding chapter/revision reference is recorded in Chapter 11.

This process ensures that comments are not only discussed but traceably processed and linked to resulting document updates.

9.6 Freeze

RFC freeze is defined as preceding Phase Omega and must be completed before implementation phases A-C begin.

Freeze is triggered by a formal governance proposal on freeze decision. The RFC is freeze-ready only when critical open decisions are resolved or reasonably deferred, no blocking inconsistencies remain between Informative and Formal chapters, prioritization and phase logic are stable, audit-gate logic per phase is complete, and stakeholder feedback plus scope and decision-log updates are cleanly integrated. Freeze output is a published “frozen” reference version; changes to frozen core decisions are only allowed via new RFC revision.

9.7 Change Control After Freeze

After freeze, a two-level change model applies. Editorial changes without impact on target state, gate logic, or scope may be applied directly. Content changes affecting target states, phase logic, gate criteria, risk assumptions, or migration principles require new RFC revision with formal governance approval. Each content change must explicitly reference impacted chapters at least through target state, roadmap phase, audit gate, and decision-log entry.

9.8 Decision and Escalation Logic During Implementation

Even after freeze, governance decisions remain required, especially for critical audit findings needing risk acceptance, gate blockers that trigger replanning, external PQ developments affecting assumptions, and deviations from defined cutover/security principles. Escalation logic is strict: first technical clarification and documentation, then explicit governance decision with Go/No-Go or deferral. Silent continuation with unresolved critical deviations is excluded.

Decision authority in this escalation path is explicitly split: - Process-level preparation and escalation coordination is handled by the RFC Governance Authority. - Binding chain-level direction and risk-acceptance decisions are made by Terra Classic Chain Governance through formal on-chain governance mechanisms.

For Phase A5 IBC migration work, execution ownership and reporting cadence must be explicitly assigned and published. At minimum, a named owner set (technical lead, governance liaison, and operations liaison) and a recurring status cadence must be defined to track counterparty outreach, relayer readiness, fallback-mode status, and unresolved blockers through to A6.

9.9 Communication Framework

Communication follows the principle “one process, two categories”: Informative communication focuses on understandable purpose, risk, and impact for community/users; Formal communication focuses on precise decision basis, gate status, and residual risk. For every phase-relevant decision, communication must include at least decision and scope, technical rationale, affected phases/gates, and open residual risks with next review point.

For counterparty-chain coordination in Phase A5, communication includes a standardized governance package template set so outreach can be translated into executable counterpart actions with minimal friction. The template set includes at least: technical change summary, risk statement, suggested sequencing model, relayer/operator checklist, and contact/escalation points.

9.10 Link to Logs

This chapter defines process; Chapter 10 is the operational decision register and Chapter 11 is the operational comment register. Every relevant governance step must create a traceable status change in the Decision Log (open, closed, deferred) including short rationale. Every submitted comment must be tracked in the Comment Log (accepted, rejected) including short rationale and, when applicable, document-change references.

10 Decision Log (Formal)

This chapter is the operational decision register of the RFC. It tracks closed, open, and deferred points in a form directly usable for governance, audit, and phase steering. The Decision Log is updated with each relevant RFC revision and is the binding reference for status transitions.

10.0.1 Q-01

Question: Which cryptographic profile is selected as the target profile for the consensus path?

Decision: ML-DSA is selected as the target profile for the consensus path.

Rationale: The consensus path is the highest-criticality path in this RFC. A single explicit target profile reduces ambiguity in architecture, testing, migration planning, and audit scope. This avoids fragmented implementation assumptions across phases and creates a clear baseline for gate-based validation.

Status: closed

Reference: Chapters 5, 6

10.0.2 Q-02

Question: Should the consensus path run in permanent hybrid mode or move to a clear PQ-only target state after cutover?

Decision: No permanent hybrid operation in consensus; the target state is a clear cutover to PQ-only operation.

Rationale: Permanent hybrid operation would increase rule complexity, increase attack surface, and make deterministic validation and incident handling harder in the most security-sensitive path. A single post-cutover signature regime is easier to reason about, test, audit, and operate safely.

Status: closed

Reference: Chapters 5, 6

10.0.3 Q-03

Question: Which cutover mechanism is required to preserve validator identity continuity during consensus migration?

Decision: Existing validators must be bound to new PQ consensus keys before cutover, and a voting-power threshold must be satisfied before halt/restart.

Rationale: Safe restart requires unambiguous continuity from existing validator identities to post-cutover PQ keys. Binding plus threshold gating reduces the risk of identity ambiguity, incomplete participation, and unsafe restart conditions, and provides measurable readiness criteria before operational execution.

Status: closed

Reference: Chapters 6, 8

10.0.4 Q-04

Question: What is the target transition model for wallet/tx signatures during PQ introduction?

Decision: The wallet/tx path adopts a hybrid signature model with staged rollout rather than an early hard PQ-only cutover.

Rationale: The wallet/tx path is ecosystem-dependent and includes diverse operational actors with different readiness levels. A staged hybrid model enables earlier participation by advanced actors (for example exchanges and custodians) while preserving operational continuity for slower-moving integrations and end-user tooling.

Status: closed

Reference: Chapters 5, 6

10.0.5 Q-05

Question: Should Terra Classic depend on external wallet/client release cycles for PQ support or establish its own PQ-capable client stack?

Decision: Terra Classic will build a native wallet/explorer ecosystem with explicit PQ compatibility as a long-term

target state.

Rationale: Exclusive dependence on third-party release priorities increases strategic and operational risk for user-facing adoption. A native stack improves control over roadmap alignment, release responsiveness, operational ownership, and long-term continuity of PQ-capable user infrastructure.

Status: closed

Reference: Chapters 5, 6

10.0.6 Q-06

Question: Which formal prerequisite must be met before implementation phases A-C can start?

Decision: A formal RFC freeze in preceding Phase Omega is required as start authorization before phases A-C.

Rationale: A mandatory freeze step ensures unresolved conflicts are handled before implementation, preserves scope integrity, and establishes a traceable baseline for downstream delivery and auditing. This reduces rework and prevents silent scope drift under execution pressure.

Status: closed

Reference: Chapters 6, 9

10.0.7 Q-07

Question: Is re-genesis currently accepted as a robust fallback migration strategy for Terra Classic?

Decision: No. Re-genesis is currently treated as an operational dead end and not as a robust fallback strategy.

Rationale: For current state size and operational constraints, no robust proof exists that safe, reproducible export/import can be completed in acceptable and reliable conditions. Depending on such an unproven fallback would shift migration risk into an unvalidated emergency path.

Status: closed

Reference: Chapter 8

10.0.8 Q-08

Question: What minimum gate discipline is mandatory between roadmap phases?

Decision: The defined gate matrix is mandatory, including blocker classes, pass criteria, and re-audit rules.

Rationale: Without binding gate discipline, risk can leak from prototype and migration work into production transitions. A formal gate baseline establishes objective continuation criteria, creates governance traceability, and enforces consistent risk treatment across phases.

Status: closed

Reference: Chapter 7

10.0.9 Q-09

Question: How exactly should the wallet/tx hybrid path be specified regarding key/account types, compatibility windows, and activation logic?

Decision: No final decision yet. Working options include conservative entry, staged activation, and aggressive rollout profiles.

Rationale: The decision depends on balancing security requirements, integration effort, user-experience risk, and operational complexity across heterogeneous ecosystem actors. Finalization requires B1-level architecture and compatibility evidence.

Status: open

Reference: Chapters 5, 6, target phase B1

10.0.10 Q-10

Question: Which formal trigger set should govern consensus cutover Go/No-Go decisions beyond minimum threshold criteria?

Decision: No final decision yet. Working options include strict criteria sets, weighted decision models, and multi-stage approval logic.

Rationale: Minimum thresholds alone may not capture full operational readiness and residual risk conditions. The final trigger model must preserve safety/liveness, deterministic behavior, and practical operational controllability in high-pressure cutover conditions.

Status: open

Reference: Chapters 6, 8, target phase A4-A6

10.0.11 Q-11

Question: Which governance model should define the relationship between Terra Classic governance and the Phase C carrier entity?

Decision: No final decision yet. Working options include direct governance binding, mandated sub-structure, and hybrid governance models.

Rationale: The relationship must ensure accountability and transparency while preserving operational ability to act and reliable control of critical assets (for example DNS, release accounts, distribution channels). The model must be precise enough to avoid governance ambiguity during incidents.

Status: open

Reference: Chapters 6, 9, target phase C2

10.0.12 Q-12

Question: Which additional re-audit regime applies if B3 and B4 are merged into one execution block?

Decision: No final decision yet. Working options include trigger-based mandatory re-audits and case-by-case governance decisions.

Rationale: Merging B3/B4 may improve delivery flow but can compress review separation between implementation and migration logic. The final regime must preserve security assurance without creating disproportionate process overhead or timing risk.

Status: open

Reference: Chapters 6, 7, target phase B3-B4

10.0.13 Q-13

Question: How should external PQ monitoring and alignment be operationalized in cadence and decision impact?

Decision: No final decision yet. Working options include periodic cycles, event-driven triggers, and combined operating models.

Rationale: The mechanism must be responsive to external evidence changes without overwhelming governance and technical teams with noise. It must define when new external findings trigger review-only actions versus binding RFC or implementation adjustments.

Status: open

Reference: Chapters 6, 9, target phase C3

10.0.14 Q-14

Question: Should a detailed re-genesis emergency specification be finalized now as part of this RFC baseline?

Decision: Deferred. No final specification is adopted in the current RFC baseline.

Rationale: A robust and reproducible technical proof for safe large-state export/import is still missing. The decision is deferred until credible end-to-end evidence under realistic operating conditions is available.

Status: deferred

Reference: Chapters 8, 9

10.0.15 Q-15

Question: Should the RFC define a final post-ML-DSA long-term cryptographic profile at this stage?

Decision: Deferred. No final post-ML-DSA profile is fixed in the current RFC revision.

Rationale: External PQ standards and cryptanalysis are still evolving. Prematurely freezing a long-term successor profile would risk locking the roadmap to assumptions that may change with new evidence and standardization

outcomes.

Status: deferred

Reference: Chapters 5, 9

10.0.16 Q-16

Question: Should the RFC feedback process run in fixed iterative windows and formal review rounds, or as a continuous open feedback process?

Decision: The RFC feedback process is continuous during the open phase and not organized by fixed iterative windows or mandatory review rounds.

Rationale: In Terra Classic's decentralized environment, continuous intake and ongoing incorporation of comments provides better operational fit than institutional round cycles. Comments may trigger single revisions or be consolidated into combined revisions, provided per-comment traceability remains explicit in the Comment Log and corresponding decision updates remain explicit in the Decision Log.

Status: closed

Reference: Chapters 9, 11

10.1 Status Scheme and Transition Rules

Status values: - open: question is decision-ready but not finally decided. - closed: question is finally decided and currently binding. - deferred: question is intentionally postponed and linked to a clear re-trigger.

Status transitions: - open -> closed only with documented governance decision and chapter reference. - open -> deferred only with reasoned deferral rationale and explicit re-trigger. - deferred -> open once defined re-trigger occurs. - Reversal of a closed direction decision only via RFC revision under freeze/change-control rules from Chapter 9.

11 Comment Log (Formal)

This chapter is the operational register for RFC comments. It tracks each submitted comment and whether it was accepted or rejected by editors. The Comment Log is updated with each relevant RFC revision and is the binding reference for comment-processing traceability.

11.1 Submission Channel

Comments are submitted via RFC mailing list (to-be-done), as defined in the RFC process chapter (Chapter 9). Operational details of the list (address, archive policy, moderation rules) are published for the open RFC phase and kept current during continuous feedback intake.

11.2 Processing Rule

For each submitted comment, editors perform a documented review and assign one status:

- accepted: comment is accepted for incorporation.
- rejected: comment is not incorporated.

Accepted comments usually result in document changes. In those cases, the resulting chapter/revision reference must be recorded in this log.

11.3 Comment Entries

11.3.1 C-001

- **Date:** to-be-done
- **From:** to-be-done
- **Submission channel:** mailing list (to-be-done)

- **Chapter reference:** to-be-done
- **Comment summary:** to-be-done
- **Editor review outcome:** accepted | rejected
- **Short rationale:** to-be-done
- **Resulting document change reference:** to-be-done (required when accepted)

11.4 Status and Traceability Rules

Every submitted comment must receive exactly one final outcome, either accepted or rejected. Each outcome requires a short rationale so that the editorial decision remains traceable. If a comment is marked as accepted, a document-change reference is mandatory, including chapter and revision reference. Editorial wording cleanups may be grouped in one document change, but each accepted comment still needs an explicit mapping in this log.